StoX

Indian Equities Analysis PWA

Technical Specification

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Comprehensive documentation of all technical indicators, entry score system, exit score system, scoring calculations, multi-timeframe analysis, classification thresholds, and modifier logic used in the StoX analysis platform.

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1. Overview

1.1 System Architecture

StoX is a Progressive Web App (PWA) for Indian equities analysis, built with vanilla React 18 and loaded via Babel standalone transpilation (no build step). All JavaScript files are loaded via script tags in index.html.

The system analyzes 200 NIFTY stocks across multiple timeframes (Daily, Hourly, Weekly) using 50+ technical indicators organized into a hierarchical scoring system.

1.2 Scoring Philosophy

The scoring system uses a three-pillar model for entry signals and a four-pillar model for exit signals. Each pillar aggregates related indicators into a composite score, providing a structured approach to technical analysis.

Key principles:

- Multi-timeframe confirmation: Higher timeframes (Weekly) provide context, while lower timeframes (Hourly) provide timing
- Volatility normalization: ATR percentile ranks adapt thresholds to current market conditions
- Pillar ceilings: Each pillar has a maximum score, preventing any single indicator from dominating
- Modifiers: Bonuses and penalties adjust scores based on market structure and risk factors

2. Technical Indicators

2.1 Trend & Moving Averages

Indicator	Description	Parameters	Formula
SMA	Simple Moving Average	period=20	Sum of closes / period
EMA	Exponential Moving Average	period=20	alpha=2/(period+1), EWM
WMA	Weighted Moving Average	period=20	Weighted sum / period*(period+1)/2
HMA	Hull Moving Average	period=20	WMA(2*WMA(n/2) - WMA(n), sqrt(n))
KAMA	Kaufman Adaptive MA	period=10	Adapts smoothing based on efficiency ratio
VWAP	Volume-Weighted Avg Price	period=10	TP*Volume / Volume cumulative
Mansfield RS	Relative Strength vs index	n=52	(stock/index*100)/SMA-1)*100
Beta	Sensitivity to index	n=60	Rolling OLS regression slope

2.2 Trend Direction & Strength

Indicator	Description	Parameters	Formula
ATR	Average True Range	period=14	EWM of True Range
ADX	Trend strength (0-100)	period=14	Returns adx, plusDI, minusDI
SuperTrend	Trend direction + stop	period=10, mult=3	ATR-based bands
PSAR	Parabolic SAR	af=0.02, step=0.02	SAR accelerates toward price
Vortex	Trend direction	period=14	Returns plus, minus
Aroon	Time since extreme	period=25	Returns up, down, osc
Ichimoku	Multi-component cloud	various	Tenkan(9), Kijun(26), etc.
MACD	Trend momentum	12,26,9	macd, signal, histogram
TSI	True Strength Index	25,13	Double-smoothed momentum
STC	Schaff Trend Cycle	23,50,10	Stochastic of MACD
AO	Awesome Oscillator	-	SMA(5) - SMA(34) of midpoints

2.3 Momentum Oscillators

Indicator	Description	Parameters	Formula
RSI	Overbought/oversold	period=14	100-100/(1+avgGain/avgLoss)
StochRSI	RSI of RSI	14,3,3	Returns k, d
Williams %R	Distance from high	period=14	(HH-close)/(HH-LL)*-100
CCI	Price vs mean deviation	period=20	(TP-mean)/(0.015*MAD)
ROC	Rate of change	period=12	(close-close_n)/close_n*100
Momentum	Absolute change	period=10	close-close_n
MFI	Money Flow Index	period=14	RSI-like, volume-weighted
CMF	Chaikin Money Flow	period=20	Cumulative CLV*volume/volume
FI	Force Index	period=13	EWM of (change*volume)

2.4 Volume-Based Indicators

Indicator	Description	Parameters	Formula
OBV	On-Balance Volume	-	Cumulative volume by direction
PVT	Price Volume Trend	-	OBV weighted by % price change
KVO	Klinger Volume Osc	34,55,13	Volume flow by trend direction
Anchored VWAP	VWAP from anchor	anchor=252 bars	TP*Vol/Vol cumulative
VP	Volume Profile	24 bins, 60 bars	Returns POC, VAH, VAL
TTM Squeeze	Volatility squeeze	BB20,2, KC1.5	True when BB inside KC
Accum/Dist	Accum vs distribution	-	CLV-based ADL

2.5 Structure & Volatility

Indicator	Description	Parameters	Formula
BB	Bollinger Bands	20,2	SMA +/- 2*StdDev
KC	Keltner Channels	20,1.5	EMA +/- 1.5*ATR
DC	Donchian Channels	20	Rolling highest high / lowest low
Chandelier	Trailing stop	22,3	Highest high - 3*ATR
Darvas Box	Breakout box	20,3	Confirmed swing high
Fibonacci	Retracement levels	50 bars	0.236,0.382,0.5,0.618,0.786
Pivot	Support/resistance	prev H/L/C	Classic + Camarilla levels
ZigZag	Trend pivots	5%	Filters noise < threshold
Fractals	Swing points	5-bar	Up fractal = bar high max of 5
Choppiness	Trend vs range	14	<38.2 trending, >61.8 ranging
MTF Align	Multi-TF trend	-	EMA9/21/50/SMA100/200 score

3. Entry Score System

3.1 Three-Pillar Model Overview

The entry score system uses a three-pillar model that aggregates 50+ indicators into three composite scores:

```
Raw Total = Trend Health (0-30) + Pullback Quality (0-30) + 4% Probability (0-40)
```

```
Final Score = clamp(0, 100, Raw Total + Modifiers)
```

Each pillar measures a different aspect of the entry setup:

- Trend Health: Is the stock in a confirmed uptrend? (30 points max)
- Pullback Quality: Is the stock at a favorable entry point? (30 points max)
- 4% Probability: How likely is a 4% move from current price? (40 points max)

3.2 Pillar 1: Trend Health (30 pts)

Trend Health measures the strength and quality of the uptrend. Higher scores indicate stronger trends with multiple confirmations.

Component	Points	Condition	Meaning
Price > SMA(50)	5	close > sma50	Stock above 50-day average
SMA(20) > SMA(50)	5	sma20 > sma50	Golden cross structure
Price > SMA(20) or VWAP	5	close > sma20 OR close > vwap	Short-term support held
ADX + DI	5	adx >= 25 AND +DI > -DI	Strong directional trend
Mansfield RS	5	rsMansfield > 0	Outperforming Nifty50
MACD Cross	5	macd > signal	Bullish momentum
Weekly HA Bullish	2.5	weeklyHABullish === true	Higher TF confirms uptrend
SMA(20) Slope	2.5	sma20Slope5 > 0 AND close > sma20	Rising 20-day average

3.3 Pillar 2: Pullback Quality (30 pts)

Pullback Quality measures whether the stock is at a favorable entry point. Higher scores indicate better risk/reward setups with consolidation near support.

Component	Points	Condition	Meaning
Distance (inner)	10	+/- 1.0 ATR	Price near support (tight)
Distance (outer)	5	+/- 1.5 ATR	Price near support (wider)
Candle Color	5	close > open	Bullish candle
BB Width Squeeze	5	bbWidth < bbWidthPrev5	Volatility contracting
RSI Oversold	5	StochRSI<20 OR RSI<40/35	Oversold condition
Volume Confirm	5	volRatio > 1.5 AND close > open	Above-avg volume on up day

3.4 Pillar 3: 4% Probability (40 pts)

4% Probability measures how likely the stock is to achieve a 4% gain. Higher scores indicate better probability based on volatility, target distance, and efficiency.

Component	Points	Condition	Meaning
Target Reachable T1	15	> 2.0 ATR	4% move fits in < 2 ATR (high prob)
Target Reachable T2	12	> 1.5 ATR	4% move fits in ~2.67 ATR
Target Reachable T3	8	> 1.0 ATR	4% move fits in ~4 ATR
Target Reachable T4	3	<= 1.0 ATR	4% move needs > 4 ATR (low prob)
Target Distance T1	10	0.25-2.0 ATR	Sweet spot: close enough, far enough
Target Distance T2	5	0-3.0 ATR	Extended range
Vol Sweet T1	10	ATR percentile 30-70%	Moderate volatility
Vol Sweet T2	5	ATR percentile 20-80%	Broader moderate zone
Efficiency Ratio	5	> 0.4	Price trending efficiently

3.5 Modifiers

Modifiers adjust the entry score based on market structure and risk factors. Bonuses reward favorable conditions, while penalties penalize risky setups.

Modifier	Amount	Condition	Meaning
Low Beta + Low Vol	-10	beta < 0.5 AND atrPercentile < 25%	No expansion expected
Spike Day	-10	gapPct > 3% OR detectSpike	Blow-off top risk
Stability Risk	-15	stability20 < 0.3	Erratic price action
MTF Alignment	+10	D + W scores >= 65	Multi-timeframe confirmation
High Vol Bonus	+5	atrPercentile > 80 AND er > 0.5	Strong trend in high vol

Score Bounds:

```
Max Positive Modifiers: +10 (MTF) + 5 (High Vol) = +15
```

```
Max Negative Modifiers: -10 (Low Beta) + -10 (Spike) + -15 (Stability) = -35
```

```
Theoretical Range: -35 to 115, but capped to 0-100
```

3.6 Classification System

Entry scores are classified into five categories that determine position sizing and alerts:

Classification	Threshold	Allocation	Color	Description
STRONG_BUY	>= 80	100%	Green	Strong bullish setup, full position
BUY	>= 65	70%	Green	Good setup, standard position
WATCHLIST	>= 50	40%	Yellow	Potential setup, reduced position
NEUTRAL	>= 35	0%	Purple	No clear signal, avoid
AVOID	< 35	0%	Red	Weak setup, do not enter

4. Multi-Timeframe Scoring

4.1 Timeframe Weights

When multiple timeframes are available, scores are aggregated using configurable weights:

Timeframe	Weight	Decimal	Role
Daily (D)	55%	0.55	Primary decision timeframe
Hourly (H)	30%	0.30	Entry timing and short-term momentum
Weekly (W)	15%	0.15	Higher timeframe context

4.2 Aggregation Logic

Each pillar is aggregated independently across timeframes:

```
Aggregated Pillar = (D_score * D_weight + H_score * H_weight + W_score * W_weight) / wSum
```

Where wSum is the sum of weights for available timeframes. If a timeframe is missing, its weight is redistributed proportionally to the others.

The aggregated pillar is then capped at its maximum (30/30/40) to prevent over-scoring.

4.3 MTF Alignment Bonus

A +10 bonus is awarded when both Weekly and Daily raw scores are >= 65, indicating strong multi-timeframe confirmation.

```
MTF Bonus = +10 IF (Weekly_raw >= 65 AND Daily_raw >= 65)
```

This bonus only applies when using multi-timeframe scoring. It is not available in single-timeframe mode.

5. Exit Score System

5.1 Four-Pillar Exit Model

The exit score uses a four-pillar model, each measuring a different aspect of deterioration:

Pillar	Max Points	Description
Trend Breakdown	25	EMA/MACD/ADX breakdown signals
Momentum Exhaustion	25	RSI/StochRSI/CCI overbought
Volume Distribution	25	OBV/PVT/KVO divergence
Structural Breakdown	25	BB/Donchian/Ichimoku breakdown

5.2 Exit Modifiers

Exit modifiers adjust the exit score based on market conditions and position status:

Condition	Amount	Meaning
Index trend >= 65	-8	Strong market, reduce exit urgency
EMA9>EMA21 + MACD bullish	-5	Short-term trend intact
Price decline 3 bars + low vol	-6	Low conviction decline
Near support (1.5% of S1)	-5	Support may hold
Held <3 days + entry >70	-5	Too early to exit strong setup
Index trend < 35	+5	Weak market, increase exit urgency
Distribution days >= 60%	+5	Institutional selling
Price < 97% of entry	+5	Stop loss territory
Price < 98.5% of entry	+3	Approaching stop
Price < HMA16 + EMA9	+5	Short-term trend broken
Golden exit (spike near 4%)	+5	Take profit on spike
Spike toward 4% (2-3%)	+3	Lock partial profits

5.3 Exit Classification

Exit scores are classified into six action levels:

Classification	Threshold	Action
URGENT_EXIT	>= 85	Full exit immediately
EXIT	>= 70	Full exit at current price
PARTIAL_EXIT	>= 55	Exit 50%, tighten stop to 1.5x ATR
TIGHTEN_STOP	>= 40	Move stop to breakeven or 1x ATR
MONITOR	>= 25	Watch for escalation
HOLD	< 25	All conditions intact

6. Confidence Models

6.1 Session Confidence

Session Confidence answers: "Will the target hit TODAY?" It uses 15-minute intraday data, VWAP, intraday ADX, MFI, ROC, ADR headroom, and time remaining.

Score	Recommendation
>= 70	Let it ride - strong chance of tagging +4% today
>= 40	Wait & watch - keep a tight stop
< 40	Low odds - bank the gain today

6.2 Horizon Confidence

Horizon Confidence answers: "Will the target hit in N days?" It uses a statistical lognormal model with drift (hourly/daily/RS) and volatility.

Score	Classification	Meaning
>= 75	HIGH_CONVICTION	Strong statistical edge
>= 60	CONFIDENT	Good probability
>= 45	MODERATE	Average probability
>= 30	LOW_CONVICTION	Below average
< 30	UNFAVORABLE	Low probability

6.3 Forward Confidence

Forward Confidence is a 5-day or 10-day wrapper that estimates the probability of reaching +4% within the forward period. It is used in the screener for stock ranking.

7. Configuration

7.1 SCORE_CONFIG Parameters

All scoring parameters are centralized in SCORE_CONFIG, which is persisted to localStorage and can be modified via the UI. The configuration is organized into sections:

Section	Description
pillarMax	Maximum scores for each pillar (30/30/40)
tfWeights	Timeframe weights (D=0.55, H=0.30, W=0.15)
trendHealth	11 parameters for Trend Health pillar
pullbackQuality	12 parameters for Pullback Quality pillar
prob4	22 parameters for 4% Probability pillar
modifiers	12 parameters for modifiers
classification	4 thresholds for entry classification

7.2 Configurable Thresholds

Key configurable thresholds in SCORE_CONFIG:

Parameter	Default	Description
adxThreshold	25	ADX level for trend confirmation
mansfieldRSThreshold	0	Minimum RS to outperform index
sma20SlopeThreshold	0	Minimum slope for rising SMA
stochRSIThreshold	20	StochRSI oversold level
rsiOversoldNormal	40	RSI oversold in normal vol
rsiOversoldHighVol	35	RSI oversold in high vol
volRatioThreshold	1.5	Volume confirmation multiplier
targetATR_threshold1	2.0	High probability ATR threshold
targetATR_threshold2	1.5	Medium probability threshold
targetATR_threshold3	1.0	Low probability threshold
efficiencyRatioThreshold	0.4	Minimum ER for trending
mtfAlignThreshold	65	Score for MTF alignment bonus
spikeGapThreshold	3	Gap % for spike detection
stabilityThreshold	0.3	Stability floor for penalty

Appendix A: Key Formulas

A.1 ATR Percentile Rank

ATR Percentile Rank determines where current volatility sits relative to its 100-bar history:

```
atrPercentile = (count of bars where ATR% < current ATR%) / 100 * 100
```

A.2 Stability Score

Stability Score measures price consistency over the lookback period:

```
stability = 0.6 * (positive_bars / total) + 0.4 * (1 - CV/2)
```

Where CV is the coefficient of variation of returns. Higher values indicate steadier price action.

A.3 Efficiency Ratio

Efficiency Ratio measures how directly price moves toward its target:

```
ER = |close - close_n| / sum(|close_i - close_i-1|) for i=1 to n
```

Values close to 1 indicate efficient trending; values near 0 indicate choppy price action.

A.4 Mansfield Relative Strength

Mansfield RS measures relative strength vs its own 52-week moving average:

```
RS = (stock/index * 100) / SMA(stock/index, 52) - 1) * 100
```

Positive values indicate the stock is outperforming its own historical relative strength.